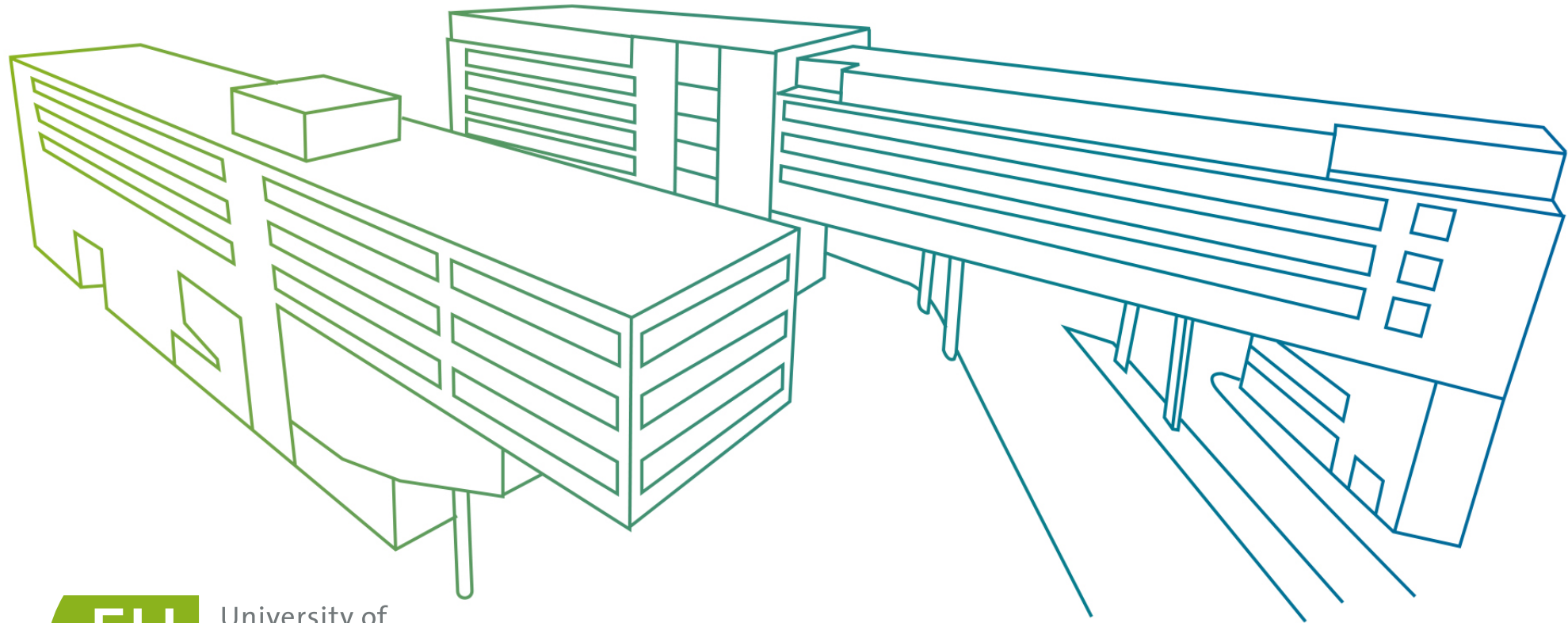




Web Technologies

HTML Forms

New requirement

- Remember the UAS Technikum Wien site from our first lesson?
- The stakeholders have decided they want a newsletter registration form
- The user should be able to enter their email address
- There should also be a checkbox for accepting the privacy policy
- Both fields must be filled in
- There should also be a button to submit the form

The graphical mockup is presented on the next slide:

The mockup

Newsletter Registration

Email *

☐ Yes, I would like to receive the newsletter and agree to the [Privacy Statement](#)



I'm not a robot



reCAPTCHA
Privacy - Terms

SUBMIT

Start with text again

Newsletter Registration

Email *

Yes, I would like to receive the newsletter
and agree to the Privacy Statement zu. *

SUBMIT

- The words "Privacy Statement" should link to the statement
- How would the HTML look like? (Invent elements if you don't know the real ones)

What we know

```
<h1>Newsletter Registration</h1>
<form?>
  <email?>Email *</email?>
  <checkbox?>
    Yes, I would like to receive the newsletter and agree to the
    <a href="/privacy">Privacy Statement</a>
    zu. *
  </checkbox?>
  <button?>SUBMIT</button?>
</form?>
```

What did we get right?

Input

- `<form>` is correct!
- `<button>` is partly correct, but we need to add an attribute so that it functions as a submit button.
 - `type="submit"`
- There is no `<email>` or `<checkbox>` element, but rather a generic `<input>` element
 - It has a `type` attribute for the different input types
 - And it is a void element

Let's take a look at the updated version:

<input>

```
<h1>Newsletter Registration</h1>
<form>
  <!-- Email * -->
  <input type="email">
  <!-- Yes, I would like to receive the newsletter and agree to the
        <a href="/privacy">Privacy Statement</a>
        zu. * -->
  <input type="checkbox">
  <button type="submit">SUBMIT</button>
</form>
```

- But if `<input>` is a void element, where can I add the texts and the link to the privacy policy?

<label>

- Every `<input>` needs a corresponding `<label>`

```
<h1>Newsletter Registration</h1>
<form>
  <label>Email *</label>
  <input type="email">
  <input type="checkbox">
  <label>
    Yes, I would like to receive the newsletter and agree to the
    <a href="/privacy">Privacy Statement</a>
    zu. *
  </label>
  <button type="submit">SUBMIT</button>
</form>
```


Are we finished?

- Not yet!
- In HTML forms, every label needs a corresponding label
- We did that, but how does the browser know which label belongs to which input?

```
<label>Email *</label>
<input type="email">
<input type="checkbox">
<label>
  Yes, I would like to receive the newsletter and
  agree to the <a href="/privacy">Privacy Statement</a> zu. *
</label>
```

Identification

- Every HTML element *can* have the `id` attribute
- This is used to clearly identify one singular element in your document
- It has to be unique in the document!
- The `<label>` can reference the element using the `for` attribute

How would that look like?

The final form?

```
<h1>Newsletter Registration</h1>
<form>
  <label for="email">Email *</label>
  <input id="email" type="email">
  <input id="privacy" type="checkbox">
  <label for="privacy">
    Yes, I would like to receive the newsletter and agree to the
    <a href="/privacy">Privacy Statement</a>
    zu. *
  </label>
  <button type="submit">SUBMIT</button>
</form>
```

One additional thing

- We *need* an email address and the checkbox *needs* to be checked
- Those fields are not optional
- You could say, the input is `required`

```
<input id="email" type="email" required>  
<input id="privacy" type="checkbox" required>
```

- `required` is a boolean attribute, it doesn't have a value (`= "value"`), it is either present or absent

What we have

```
<h1>Newsletter Registration</h1>
<form>
  <label for="email">Email *</label>
  <input id="email" type="email" required>
  <input id="privacy" type="checkbox" required>
  <label for="privacy">
    Yes, I would like to receive the newsletter and agree to the
    <a href="/privacy">Privacy Statement</a>
    zu. *
  </label>
  <button type="submit">SUBMIT</button>
</form>
```

There is more!

- Remember: HTML is all about structure and content!
- We have the structure for input
- We have the content to display the meaning to the user
- What we are missing is a way to process the data
 - Where can we send the data?
 - Are there different ways of sending data?
 - This will be part in one of the upcoming lessons

Key Takeaways

Essential Form Structure: Forms use the `<form>` element as a container, with `<input>` elements for user input. The `<input>` element is a void element that uses different `type` attributes to specify the input type.

Labels Are Critical: Every input field must have a corresponding `<label>` element to provide context and meaning. Labels serve both accessibility and usability purposes, telling users what each input field is for.

Key Takeaways

Connecting Labels to Inputs: Labels are connected to their corresponding inputs through the `id` and `for` attributes. Each input gets a unique `id`, and its label references that ID using the `for` attribute. This creates a semantic relationship between the label and input.

Button Functionality: Submit buttons need `type="submit"` to function properly as form submission triggers.

Boolean Attributes: `required` is a boolean attribute, that sets an input *not* optional.

Key Takeaways

Separation of Concerns: HTML focuses on structure and content, it defines what the form contains and how it's organized. The actual data processing happens separately, which reinforces HTML's role as a markup language rather than a programming language.

More to come: We are not yet finished with the HTML form. A few additional attributes still need to be set so that the actual data processing can take place.